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Course: 3BME

Exam: Modelling II MP307

MP 307

① Section A

Q1a $f(t) = \begin{cases} \frac{2t+1}{4} & 1 \leq t \leq 2 \\ 0, & 0 \leq t \leq 1, t \geq 2 \end{cases}$

① A prob density function $f(t) \geq 0$
 $\text{Prob}(1 \leq t \leq 2) = \int_1^2 f(t) dt$

with condition
 $\text{Prob}(-\infty < t < \infty) = \int_{-\infty}^{\infty} f(t) dt = 1$

①① $P_k = \text{prob}(\text{state } k \text{ occurs})$
 $k = \text{time} = t$

$$\sum_{1 \leq t \leq 2}^{\infty} \frac{2t+1}{4}$$

①① $\sigma^2 = \langle t^2 \rangle - \mu^2$
 $\mu = \langle t \rangle = \sum_{t \geq 0} t p_t$

$$= \sum_{1 \leq t \leq 2}^{\infty} \frac{2t^2+t}{4}$$

$$\langle t^2 \rangle = \sum_{1 \leq t \leq 2}^{\infty} \frac{2t^4+t^2}{4}$$

$$\sigma^2 = \sum_{1 \leq t \leq 2}^{\infty} \frac{2t^2+t}{4} - \sum_{1 \leq t \leq 2}^{\infty} \frac{2t^4+t^2}{(4)^2}$$

$$\sigma^2 = t^2 \left| 2 \right.$$

$$\sigma = \sqrt{\sigma^2}$$

③

$$① \quad 0.8\pi_1 + 0.2\pi_2 = \pi_1$$

$$② \quad 0.7\pi_2 + 0.3\pi_3 = \pi_2$$

$$③ \quad 0.5\pi_1 + 0.5\pi_3 = \pi_3$$

$$④ \quad \pi_1 + \pi_2 + \pi_3 = 1$$

$$① \quad 0.2\pi_2 = 0.2\pi_1 \\ \pi_2 = \pi_1$$

$$② \quad 0.3\pi_3 = 0.3\pi_2 \\ \pi_3 = \pi_2$$

$$③ \quad 0.5\pi_1 = 0.5\pi_3 \\ \pi_1 = \pi_3$$

$$\pi_1 = \pi_2 = \pi_3$$

$$④ \quad \pi_1(1+1+1) = 1 \quad \pi_2(1+1+1) = 1 \\ 3\pi_1 = 1 \quad 3\pi_2 = 1 \\ \pi_1 = \frac{1}{3} \quad \pi_2 = \frac{1}{3}$$

$$\pi_3(1+1+1) = 1 \\ 3\pi_3 = 1 \\ \pi_3 = \frac{1}{3}$$

Equilibrium probabilities $\Rightarrow \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$
are

$$\pi_1 = \pi_2 = \pi_3 = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$$

⑦

$$(1 - k_1 - T)(2 - T) - (-2)(-1 - k_2)$$

$$2(1 - k_1 - T) - T(1 - k_2 - T) + 2(-1 - k_2)$$

$$2 - 2k_1 - 2T - T + k_1 T + T^2 - 2 - 2k_2$$

$$T^2 - 3T + k_1 T - 2k_1 - 2k_2 = (T+1)(T+1)$$

$$T^2 + T(k_1 - 3) - 2k_1 - 2k_2 = T^2 + 2T + 1$$

$$T(k_1 - 3) = 2T \quad -2k_1 - 2k_2 = 1$$

$$k_1 - 3 = 2$$

$$-2(5) - 2k_2 = 1$$

$$-2k_2 = 11$$

$$k_1 = 5$$

$$k_2 = -\frac{11}{2}$$

Calculate the control parameter K_r

$$C(A - BK)^{-1}B$$

$$\begin{pmatrix} d & b \\ -c & a \end{pmatrix}$$

$$(0 \ 1) \frac{1}{2(1 - k_1) - (-2)(-1 - k_2)} \begin{bmatrix} 2 & 1 + k_2 \\ 2 & 1 - k_1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\Rightarrow (0 \ 1) \begin{pmatrix} 2 & 1 + k_2 \\ 2 & 1 - k_2 \end{pmatrix} = \begin{pmatrix} 2 & 1 - k_2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$= 2 \times 1 = \frac{2}{2 - 2k_1 + 2(1 - k_2)} = \frac{2}{-2k_1 - 2k_2} = K_r$$

$$K_r = -\frac{1}{k_1} - \frac{1}{k_2}$$

$$k_1 = 5 \quad k_2 = -\frac{11}{2} \quad r = 2$$

$$U = -k_1 c_1 - k_2 c_2 + 2 \left(\frac{-2}{2k_1} - \frac{2}{2k_2} \right)$$

$$= -k_1 c_1 - k_2 c_2 - \frac{2}{k_1} - \frac{2}{k_2}$$

(a)

$$\Rightarrow \pi_{k-1} = p_{k-2} \Rightarrow p_2 \pi_{k-1} = p_1 \pi_{k-2}$$

$$p_2 \pi_k = \frac{-p_1 \pi_{k-1}}{k-1} + (p_1 + p_2) \pi_{k-1}$$

$$= \pi_k = \frac{p_1}{p_2} \pi_{k-1} = p \pi_{k-1}$$

So (i) holds for k

$$\Rightarrow \pi_k = p^k \pi_0$$

We know $\sum_{k=0}^{\infty} \pi_k = 1 \Rightarrow \sum_{k=0}^{\infty} p^k \pi_0 = 1$

$$\Rightarrow \pi_0 = \frac{1}{\sum_{k=0}^{\infty} p^k} \Rightarrow \pi_0 = \frac{1}{\frac{1-p^{r+1}}{1-p}} = \frac{1-p}{1-p^{r+1}}, p \neq 1$$

$$\pi_0 = \frac{1}{r+1}, p=1$$

(b) $\alpha_k = \alpha$ $B_k = \beta$

$\alpha > 0, \beta > 0$

Let $p_k = \text{Prob}(K \text{ customers arrive in time } t = m\Delta t)$

(11)

$$C) \text{Prob}(K > 3) = 1 - \text{Prob}(K \leq 3)$$

α = arrival

βK = servicing

$\alpha = 300 \text{ per hour} = 15 \text{ per min}$
~~per min~~

$$\beta = \frac{1}{4} \text{ cust/min} = \frac{60}{4} / \text{hour} = 15 \text{ cust/hour}$$

$\alpha = 300 \text{ cust/hour}$		5 cust/min	same
$\beta = 15 \text{ cust/hour}$		$\frac{1}{4} \text{ cust/min}$	

$$\rho = \frac{\alpha}{\beta} = \frac{300}{15} = 20 > 1$$

$$\pi_3 = \frac{20^3 e^{-20}}{3!}$$

$\alpha = 300 \text{ per hour}$
 $\beta = 4 \text{ per min} = 15 \text{ per hour}$

$$\pi_3 = \frac{20^3 e^{-20}}{3!} = 2.74820 \times 10^{-9} \text{ queue size}$$

$$1 - \text{prob}(K \leq 3) = 0.9999972518$$

that the queue
is longer than 3 $\nearrow = 0.990$

(13)

$$\sigma^2 = \langle t^2 \rangle - \mu^2$$

$$\mu = \langle t \rangle = \sum_{t \geq 0} t p_t$$

$$\langle t \rangle = \sum_{t=0}^{\infty} \frac{8t^2 + 1}{4} \quad \langle t \rangle^2 = \frac{4t^2 + 1}{4}$$

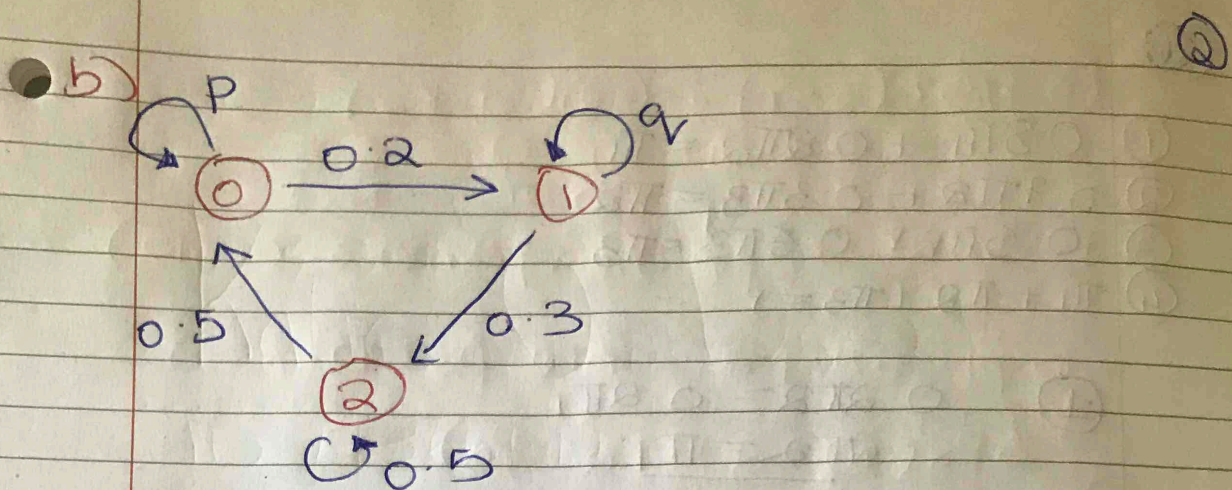
$$1 \leq t \leq 2$$

$$\sigma^2 = \sum_{1 \leq t \leq 2} \frac{4t^2 + 1}{4} - \sum_{1 \leq t \leq 2} \frac{4t^2 - 1}{16}$$

$$\sigma^2 = t^2 \left(\frac{4+1}{4} - \frac{4+1}{16} \right)$$

Standard deviation σ

$$\sigma = \sqrt{t^2 \left(\frac{5}{4} - \frac{5}{16} \right)}$$



i)
$$\begin{matrix}
 & 0 & 1 & 2 \\
 \begin{matrix} 0 \\ 1 \\ 2 \end{matrix} & \begin{pmatrix} p & 0.2 & 0 \\ 0 & q & 0.3 \\ 0.5 & 0 & 0.5 \end{pmatrix}
 \end{matrix}$$

$$p + 0.2 = 1$$

$$p = 0.8$$

$$q + 0.3 = 1$$

$$q = 0.7$$

$$P = \begin{pmatrix} 0.8 & 0.2 & 0 \\ 0 & 0.7 & 0.3 \\ 0.5 & 0 & 0.5 \end{pmatrix}$$

ii) Yes, the system is ergodic as you can travel to each state using other states i.e. there is some connection between the states where you can visit each state.

iii) Yes the Markov process does have equilibrium probabilities as all states are joined in one model



©

Re
write

④

$$\frac{dP_1}{dt} = 2P_1(20 - P_1 - P_2)$$

$$\frac{dP_2}{dt} = 3P_2(10 - P_1 - P_2)$$

①

P_1 = population size of P_1

P_2 = population size of P_2

20 = max population size of P_1

10 = max population size of P_2

2 P_1 = biotic population of P_1

3 P_2 = biotic population of P_2

⑦ $\dot{P}_1 = 0, \dot{P}_2 = 0$ - Equilibrium points

$$\dot{P}_1 = 0 \Rightarrow 2P_1(20 - P_1 - P_2) = 0$$

$$P_1 = 0 \quad 2(0)(20 - P_2) = 0$$

$$0 = 0$$

$$2P_1(20 - P_1 - P_2)$$

$$P_1 = 20 - P_2 \quad 2(20 - P_2)(20 - 20 + P_2 + P_2) = 0$$

$$2(20 - P_2)(0) = 0$$

$$0 = 0$$

Sub $P_1 = 0$ into $\dot{P}_2 = 0$

$$3P_2(10 - 0 - P_2) = 0$$

$$P_2 = 0$$

$$3(0)(10 - 0) = 0$$

$$0 = 0$$

Equilibrium point = (0, 0)

Section B

⑥

Q4
a)

$$A = \begin{pmatrix} 1 & -1 \\ -2 & 2 \end{pmatrix} \quad B = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad C = (0, 1)$$

$$w_o = C \begin{bmatrix} 0 & 1 \\ -2 & 2 \end{bmatrix}$$

$$\det(0 - (-2)) \neq 0$$

The system is observable as the rows are linearly independent & system is invertible

$$w_r = B \quad AB$$

$$\begin{bmatrix} 1 & 1 \\ 0 & -2 \end{bmatrix}$$

$$\det(-2 - 0) \neq 0$$

The system is reachable as the rows are linearly independent

⑥ $y = r = 2 \quad x_1 = -1 \quad T_2 = -1$

$$\frac{dc}{dt} = (A - BK)c + BKr \quad r$$

$$= \begin{bmatrix} 1 - K_1 & -1 - K_2 \\ -2 & 2 \end{bmatrix} c + 2 \begin{pmatrix} Kr \\ 0 \end{pmatrix}$$

$$y = Cc = (0 \ 1) = c_2$$

$$\det(A - BK - \tau I) = \begin{vmatrix} 1 - K_1 - \tau & -1 - K_2 \\ -2 & 2 - \tau \end{vmatrix}$$

⑧

with $k_1 = 5$ and $k_2 = -\frac{11}{2}$

Q2a)

① max queue size = r
 $\lambda r - \beta_0 = 0$

The probability of arrival and servicing is independent of the queue size. Only one customer can arrive or be serviced in one time interval.
 hence $\lambda r = \beta_0 = 0$

② Show in two steps

① $\pi_k = p \pi_{k-1} \Rightarrow \pi_k = p^2 \pi_{k-2} = \dots = p^k \pi_0$

② $\pi_0 = \frac{1-p}{1-p^{r+1}}$, $p \neq 1$, $\pi_0 = \frac{1}{1+r}$, $p = 1$

③ by Induction

using the transition matrix P we have
 $\pi = \pi P$, $\pi = (\pi_0, \dots, \pi_r)$

$$\pi_0 = \pi_0(1-p_1) + \pi_1 p_2$$

$$0 = \pi_0 p_1 + \pi_1 p_2$$

$$\pi_1 p_2 = \pi_0 p_1$$

$$\pi_1 = p_1 / p_2 \quad \pi_0 = p \pi_0 \quad (i) \text{ holds for } k=1$$

④ Assume $\pi_{k-1} = p \pi_{k-2}$ holds then we have $(k-1)^{\text{st}}$ element of $\pi = \pi P$ gives
 $\pi_{k-1} = \pi_{k-2} p_1 + \pi_{k+1} (1-p_1-p_2) + \pi_k p_2$

$$\Rightarrow p_2 \pi_k = -\pi_{k-2} p_1 + (p_1 + p_2) \pi_{k-1}$$

occurs in $\binom{m}{k} = \frac{m!}{k!(m-k)!}$ ⑩

$$p_k = \binom{m}{k} (\alpha \Delta t)^k (1 - \alpha \Delta t)^{m-k}$$

$$= \binom{m}{k} \frac{1}{m^k} (\alpha \Delta t)^k \left(1 - \frac{\alpha \Delta t}{m}\right)^{m-k}$$

$\Delta t \rightarrow 0 \quad m \rightarrow \infty$

$\binom{m}{k} \frac{1}{m^k} = \frac{m(m-1)\dots(m-k+1)}{m^k k!} \rightarrow \frac{1}{k!}$

$$\left(1 - \frac{\alpha \Delta t}{m}\right)^{m-k} = \left(1 - \frac{\alpha \Delta t}{m}\right)^m \left(1 - \frac{\alpha \Delta t}{m}\right)^{-k} \rightarrow e^{-\alpha \Delta t} \cdot 1$$

Thus the limit

$$p_k = \frac{(\alpha \Delta t)^k}{k!} e^{-\alpha \Delta t}$$

where $\sum_{k=0}^{\infty} p_k = 1$ holds

$\pi_k = p_k p_{k+1} \dots p_{\infty}$ ^{from} π_0 implies that

$$\pi_k = \frac{p^k}{k!} \pi_0, \quad \sum_{k=0}^{\infty} \frac{p^k}{k!} \pi_0 = 1 \Rightarrow e^p = \frac{1}{\pi_0}$$

hence $\pi_k = \frac{p^k}{k!} e^{-p}$ where $p = \frac{\alpha}{\beta}$

(10)

Q1a)

$$f(t) = \begin{cases} \frac{2t+1}{4} & 1 \leq t \leq 2 \\ 0 & 0 \leq t < 1, t > 2 \end{cases}$$

① A density function is the frequency of occurrences of states in the interval $[x, x+\Delta x]$ for small Δx approaches $f(t)dt$ as $N \rightarrow \infty$

~~the area~~

interval is $[1, 2]$ Prob $[1 \leq t \leq 2]$ & total area is unity

Ans

A prob density function $f(t) \geq 0$

$$\text{Prob}(1 \leq t \leq 2) = \int_1^2 f(t) dt$$

with condition

$$\text{Prob}(-\infty < t < \infty) = \int_{-\infty}^{\infty} f(t) dt = 1$$

(11)

$$\int_{1 \leq t \leq 2}$$

$$\frac{2t+1}{4}$$

$k = \text{time} = t$

$p_k = \text{prob}(\text{state occurs})$



⑤

$$P_2 = 10$$

$$3(10)(10-10) = 0$$

$$0 = 0$$

$$\text{Equilibrium point} = (0, 10)$$

$$\text{Sub } p_1 = 20 - p_2 \text{ into } \dot{p}_2 = 0$$

$$3P_2(10 - 20 + P_2 - P_2) = 0$$

$$P_2 = 0$$

$$3(0)(10) = 0$$

$$0 = 0$$

$$P_1 = 20 - 0 = 20$$

$$\text{Equilibrium point} = (20, 0)$$

$$\textcircled{d} \begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} 0 & b \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u$$

$$y = \begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + 0u$$

$$\textcircled{i} \quad w_o = C \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$$

$$\det = (b-1) \neq 0$$

System is observable as rows are linearly independent & system is invertible

$$\textcircled{ii} \quad w_r = B \quad AB \quad \det \begin{pmatrix} 0 & -b \\ 1 & 1 \end{pmatrix} \neq 0$$

System is reachable as rows are linearly independent as $\det \neq 0$